Kaffa: Solution Design

An AI export compliance officer for Ethiopian coffee exporters

Nov 6, 2026 · @Kidus

Summary

Kaffa does the job of an export compliance data officer at an Ethiopian coffee exporter: it finds the missing farm data that blocks each EU shipment, then chases it down.

From 30 December 2026, EU buyers must prove their coffee comes from land not deforested after 2020. Software for screening and paperwork already exists. Kaffa covers the work those tools leave to a person: working out what is missing, chasing field agents on Telegram in Amharic, re-checking their replies, and warning the exporter which shipments will miss their ship date.

The exporter stays in control. Fixed rules decide compliance status, Gemini handles planning and conversation, and a person approves anything sent to a buyer.

Why this role

The deadline is 54 days away, and the bottleneck is proving compliance, not stopping deforestation.

- **The rule:** the EU deforestation regulation applies from 30 December 2026 for large and medium operators, and 30 June 2027 for micro and small ones. The cut-off date is 31 December 2020.
- **The market:** the EU buys about 30% of Ethiopia's coffee exports, more than any other single market.
- **The gap:** an ODI Global study, as reported by Birr Metrics, finds traceability gaps in up to 57% of supply chains. It estimates up to 60% of smallholders could be excluded.
- **The disagreement:** Ethiopia's coffee authority says over 99% of farms are deforestation-free and disputes the mapping figures as out of date. An ODI consultant says the main challenge is the capacity to demonstrate compliance.

The obligation falls on the EU operator, so the Ethiopian exporter's problem is commercial: a buyer who cannot get clean data for a lot can stop buying it.

What already exists

Screening and paperwork are well served, so Kaffa uses them as inputs instead of competing with them.

Existing tool	What it does	What it leaves open
Botminds	Seven agents for EU compliance teams, including an Outreach agent that chases suppliers in their language	Named languages are French, Portuguese, Bahasa and English. No mention of Amharic, Telegram as an agent channel, or field agents
ECTMS	Ethiopia's national system: farm coordinates, supply chain tracking, risk assessment	Coverage and exporter access are not described in the sources read
TechnoServe app	Free Android app that captures farmer data and shares files over WhatsApp or Telegram	File sharing only, with no follow-up workflow
Whisp	Open-source API that analyses plots a user submits	Analysis only, with no compliance decision

Botminds is the closest match and shows a compliance team will pay for agent-led chasing. Kaffa takes the other side of that trade: the exporter in Addis Ababa, working with field agents on Telegram in Amharic. I found no direct equivalent, based on a limited search of public pages.

This role also fits Brain3's premise: an exporter facing this deadline needs a person to do the work more than another dashboard.

Role breakdown

The highest-value work is finding gaps and chasing people, because every later step depends on clean data arriving on time. Priority below reflects how directly a task unblocks a shipment.

Priority	Sub-function	What a person does today	How Kaffa handles it
1	Find gaps	Cross-checks lots against plot records	Automated
1	Validate plot data	Reviews coordinates and shapes by hand	Automated with fixed rules
1	Chase field agents	Calls and messages cooperatives, then tracks who replied	Automated, escalates to a person

Priority	Sub-function	What a person does today	How Kaffa handles it
1	Forecast readiness	Estimates which shipments will be ready	Automated
2	Screen for forest loss	Runs plots through a screening tool	Automated with satellite data
2	Hold or release lots	Decides what can ship	Automated hold, human release
2	Prepare buyer evidence	Assembles reports and files	Drafted, human approves
2	Update buyers	Emails status and evidence	Drafted, human approves
3	Monitor cleared plots	Re-checks plots for new alerts	Automated daily
Out of scope	Verify land legality	Reviews land documents	Not covered in this version
Out of scope	File the EU statement	Done by the EU operator in TRACES	Kaffa supplies the data

Legality is a real EUDR requirement, so the document pack will state that it is not covered.

What the agent handles

Kaffa runs one loop for each shipment: find what blocks it, chase the person who can fix it, re-check the reply, and repeat until the shipment is ready or a person must step in.

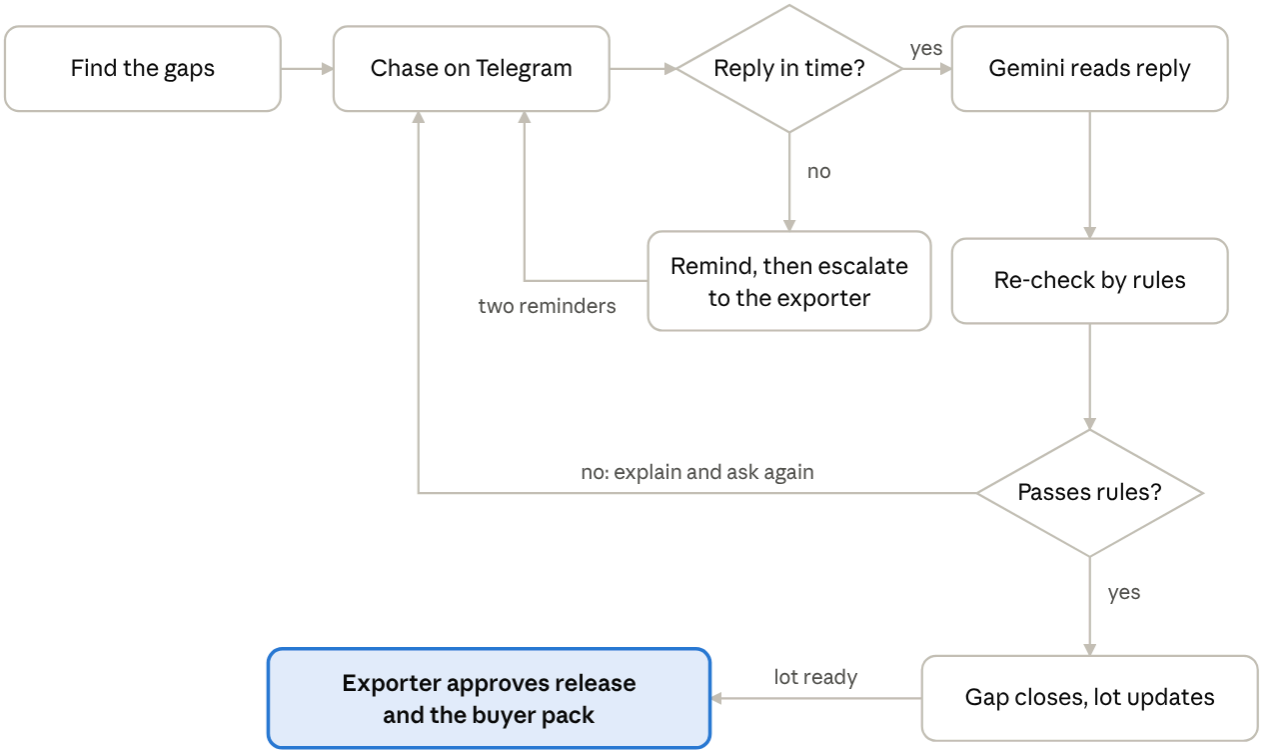
1. **Set up a shipment.** The exporter enters the buyer and ship date, then imports a cooperative's plots from CSV or GeoJSON. Each lot links to its plots, and each cooperative to a field agent.
2. **Find the gaps.** Fixed rules check each plot against the EUDR geolocation requirements: six decimal places, a polygon above 4 hectares, closed rings, and no self-intersection. They also catch missing data, duplicates and overlapping plots.
3. **Screen for forest loss.** Valid plots are checked for tree cover loss after 2020 and for recent alerts. Each plot ends as clear, review, blocked or not checked.
4. **Chase the field agent.** Kaffa groups open items by person and sends one short Telegram message, in Amharic by default, naming the exact farms and what it needs.

5. **Handle the reply.** A small plot needs one location pin. A larger plot needs pins for each corner in order, then a done message. Gemini reads free text, and the rules re-check whatever comes back.
6. **Nudge or escalate.** With no reply, Kaffa sends a reminder after 48 hours and a second after 96. It then escalates to the exporter with the blocked lot and value at stake. These timings are configurable.
7. **Forecast readiness.** Kaffa compares open items with each cooperative's recent resolution rate, then flags shipments projected to miss their ship date.
8. **Hold, split or release.** One blocked plot holds its whole lot. Kaffa can propose removing that plot, but only a person can release the lot.
9. **Prepare the buyer pack.** Once a lot is ready, Kaffa drafts a PDF summary, a GeoJSON file and a cover email for approval.

The daily run

A scheduled job runs without anyone asking. Each morning it re-checks cleared plots for new alerts, reopens affected lots, sends the nudges due that day, and updates every forecast. The exporter then gets a short brief with the three things that need attention.

A gap closes only after the rules pass the reply



chase loop · from gap to buyer pack

The highlighted step is the only one that releases a lot or sends anything to a buyer.

Trust and human control

Gemini never decides whether a plot or lot is compliant. Fixed code makes that call, so every status can be traced to a rule and a data source.

Action	Who decides	Why
Set a plot's status	Fixed rules	Results must be repeatable and auditable
Send a chase or reminder	Agent, within daily limits	Low risk, and this is the job being automated
Hold a lot	Fixed rules	Failing safe protects the buyer
Accept corrected plot data	Fixed rules, after a re-check	A reply is never trusted on its own
Clear a review plot	Exporter, with a recorded reason	Tree loss is a signal, not proof, especially in shade-grown coffee
Release a lot	Exporter	Commercial and legal responsibility stays with a person
Email a buyer	Exporter approves the draft	Outbound claims carry legal weight

Each plot gets one status. The EUDR sets no tolerance, so any detected loss needs a person's attention:

Status	Meaning	Lot effect
Clear	Valid geometry, with no loss after 2020 and no recent alert	Can ship
Review	Loss or an alert below the blocked threshold	Held until a person accepts it
Blocked	Loss covering 10% of the plot or 0.5 hectares, whichever is lower	Held until removed or disproved
Not checked	Invalid geometry, or satellite data unavailable	Held

The two thresholds are design choices, not legal figures. They stay in versioned configuration, and every check records the version used.

Three more safeguards apply:

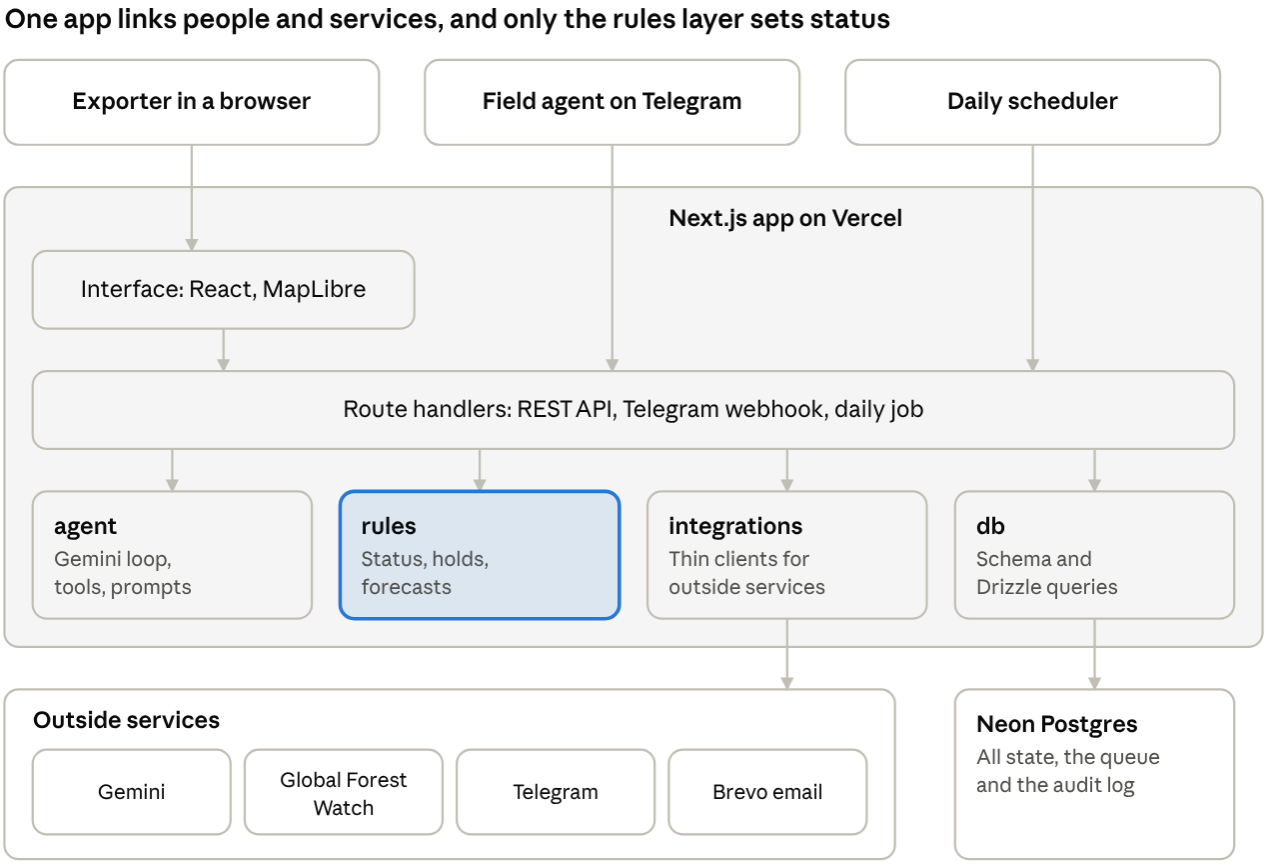
- **Untrusted replies:** a Telegram message is treated as data. Gemini can only extract fields from it, and its output goes to the rules. It has no tool that changes a status.
- **Known senders:** updates are accepted only from registered Telegram accounts, and only for that person's cooperative.
- **Audit trail:** every check, message, decision and approval is logged with its time and source.

Kaffa's output supports the buyer's due diligence. It is not a legal determination of compliance.

Real farmer names and coordinates are personal data, and Google's terms say not to submit personal information to the unpaid Gemini tier. The demo therefore uses synthetic farms only. A production version would need a paid tier.

Architecture

Kaffa is one Next.js application on Vercel, with Postgres holding all state. Keeping one deployment means Telegram webhooks and the daily job never wait on a sleeping server.



Kaffa architecture · one app, four code layers

The agent can read rule results, but it cannot change them.

Frontend

The interface uses React through Next.js, TypeScript and Tailwind, with MapLibre for the map. Pages poll the event log every two seconds, so agent activity appears live without a socket connection.

Backend

Route handlers expose the REST API, the Telegram webhook, the cron endpoint and the approval actions. The code separates four layers, each testable alone:

- `rules` : pure functions for geometry, risk status, lot holds and forecasts
- `agent` : the Gemini loop, its tools and its prompts
- `integrations` : thin clients for Gemini, Global Forest Watch, Telegram and email
- `db` : schema and queries through Drizzle

Agent loop

Gemini works through function calling. Each tool validates its arguments on the server, and failed calls return an error the model can act on.

Tool	What it does	Limit
`list_gaps`	Returns open items by shipment or person	Read only
`run_checks`	Runs rules and satellite screening on plots	Results come from code
`extract_plot_update`	Turns a reply into structured fields	Output goes to the rules
`send_chase`	Sends a Telegram message to a field agent	Registered agents, daily cap
`forecast_shipment`	Projects the ready date	Read only
`propose_lot_split`	Suggests removing a blocked plot	Needs human approval
`draft_buyer_pack`	Drafts the PDF, GeoJSON and email	Needs human approval

There is deliberately no tool for changing a status, releasing a lot or emailing a buyer. Those happen only through a signed-in person's action in the interface.

Data model

Table	Holds
`cooperatives`, `field_agents`	Who supplies the coffee, plus Telegram IDs and language
`shipments`, `lots`	Buyer, ship date, value, status
`plots`	Geometry, size, status, cooperative
`checks`	Each result, with its rule version and dataset version
`gaps`	Open items, owner, due date, reminder count
`messages`	Every Telegram message in and out
`approvals`	Drafts waiting for a person, and the decision
`events`	Audit log, which also feeds the live activity view

Satellite checks run in small batches from a queue in Postgres, which keeps each function call under Vercel's 300-second limit.

Third-party APIs and free-tier limits

Kaffa uses three third-party APIs plus email, all on free plans sized for a demo workload.

Service	Used for	Free-tier position	To confirm during build
Gemini	Planning, Amharic and English messages, reply reading	Free tier exists, limits shown in AI Studio. Ethiopia is a supported region	Best free model with function calling. The model name stays in configuration
Global Forest Watch	Tree cover loss and recent forest alerts	Free API key with polygon queries. Keys need registered domains for better rate limits	Current dataset versions and exact field names, because that guide is about 16 months old
Telegram Bot API	Chasing and receiving replies	No charge for messaging users, about one message a second per chat	Each field agent joins once through a start link

Service	Used for	Free-tier position	To confirm during build
Brevo	Buyer email and exporter alerts	Transactional send API with a verified sender	Free allowance and whether a personal address can be the sender. Gmail SMTP is the fallback
Neon	Postgres	1 GB storage, 100 compute hours, no card	None
Vercel	Hosting and daily job	300-second functions, daily cron with hourly precision	None

Scheduler decision

The daily job uses Vercel Cron instead of Google Cloud Scheduler. Kaffa needs one run a day, which the free plan covers, and this keeps the project on a single platform.

The job is a secured HTTP endpoint, so Cloud Scheduler can replace Vercel Cron with no code change if Brain3 prefers it. A manual "run now" button triggers the same code for demonstrations.

User interface

The exporter opens one screen each morning and sees which shipments are at risk and what needs a decision. The interface is responsive, so it works on a phone.

Screen	What the user sees	What the user can do
Command centre	Countdown to 30 December, shipments ranked by risk, ready date against ship date, the morning brief	Open a shipment, start the daily run
Shipment	Satellite map with plots coloured by status, lots, open gaps	Import plots, inspect a plot, start a chase
Field chase	Each field agent's open items and full Telegram conversation, with English beside the Amharic	Take over a conversation, resolve an item

Screen	What the user sees	What the user can do
Approvals	Lot releases, plot splits, review plots and buyer packs waiting for a decision	Approve, reject or edit, with a recorded reason
Activity	Every agent action as it happens, with its data source	Filter by shipment or field agent
Ask Kaffa	Chat panel on every screen	Ask questions such as "Why is the Hamburg shipment late?" or give instructions

The English translation matters. An exporter must be able to audit what the agent said on their behalf.

Failure handling, testing and demo honesty

When something fails, Kaffa fails closed: a plot stays not checked and a lot stays held. It never shows a pass it did not earn.

Failure	Behaviour
Satellite data unavailable or rate limited	Plot stays not checked and retries in the next batch
Gemini unavailable or over quota	Checks and holds continue, and chases use fixed Amharic and English templates
Telegram send fails	Retry, then escalate to the exporter
Duplicate Telegram update	Ignored, using the update ID
Reply cannot be understood	One clarifying question, then escalate after a second failure
Corrected coordinates fail the rules	The agent explains the problem and asks again
Daily job missed	The next run catches up, and the screen shows the last successful run

Testing

- **Unit tests:** geometry rules, risk status, lot holds and forecasts, written before the code they cover

- **Integration tests:** recorded Global Forest Watch responses, plus Telegram webhook handling
- **Live checks:** one real satellite query, and one real Telegram round trip with a test account
- **End to end:** a browser run of the full flow, from import to approved email

Demo honesty

- Farms, cooperatives and buyers are synthetic and labelled as such.
- Satellite results are real, because synthetic plots are placed at real locations.
- New forest alerts cannot appear on cue, so the daily-run demo uses a clearly labelled test alert.
- A native speaker checks the Amharic templates before the demo.

Scope and build order

The build order keeps one complete path working early, so anything left unfinished is an extra.

1. Data model, rules and tests
2. Global Forest Watch client and synthetic demo data
3. Telegram connection and chase loop
4. Gemini agent and forecast
5. Interface: command centre, shipment map, field chase
6. Approvals, buyer pack and email
7. Daily run and test alert
8. Deployment, live checks and demo recording

Done means: a reviewer can watch a shipment go from blocked to ready, driven by a real Telegram conversation and real satellite checks, ending in an approved buyer email.

If time runs short, cut in this order: Ask Kaffa, plot splitting, the PDF (the GeoJSON and email stay), then the activity screen as a separate page. The path from import through chase, re-check, forecast and approved email stays.

Not in this version

- Land legality checks and document review
- Direct filing to the EU's TRACES system
- Integration with the national ECTMS
- Multi-company accounts and billing

- Sign-in beyond a single exporter account